

4.3.2 Radiation and risk

Ionising radiations include some types of electromagnetic radiation and particles emitted from radioactive atoms. The risks from exposure to ionising radiation can be overestimated in some contexts but underestimated in others. This matters because ionising radiation can damage living

cells in ways that lead to the development of malignant tumours. Understanding of the properties of the different types of ionising radiation helps people to protect themselves and avoid unnecessary exposure to risk.

4.3.2.1 Absorption and emission of radiation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that the arrangements of electrons in atoms may change with absorption or emission of electromagnetic radiation.	When atoms gain energy by heating, from electricity, or by absorbing electromagnetic radiation, some electrons jump to higher energy levels. Electromagnetic radiation is given out when the electrons drop back to lower levels. The frequency of the radiation depends on the size of the energy jump. Atoms of elements such as neon and sodium give out light in the visible region of the spectrum. Other atoms, such as mercury atoms, give out light in the ultraviolet region.	WS 1.2 Use of the energy level model of the atom.

4.3.2.2 Radioactive decay

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that some nuclei are unstable and may emit alpha particles, beta particles, or neutrons, and electromagnetic radiation as gamma rays; relate these emissions to possible changes in the mass or the charge of the nucleus, or both. Use names and symbols of common nuclei and particles to write balanced equations that represent radioactive decay.	The nuclear radiation emitted may be: • an alpha particle (α) – this consists of two neutrons and two protons; it is identical to the nucleus of a helium atom • a beta particle (β) – a high-speed electron ejected from the nucleus as a neutron turns into a proton • a gamma ray (γ) – electromagnetic radiation from the nucleus • a neutron (n). Nuclear equations are used to represent radioactive decay. In a nuclear equation an alpha particle may be represented by the symbol: ${}^4_2\text{He}$ and a beta particle by the symbol: ${}^0_{-1}\text{e}$ The emission of the different types of ionising radiation may cause a change in the mass and/or the charge of the nucleus. For example, alpha decay causes the atomic number to decrease by two units and the mass number by four units: ${}^{219}_{86}\text{Ra} \longrightarrow {}^{215}_{84}\text{Po} + {}^4_2\text{He}$ There is no change in mass number during beta decay but the atomic number increases by one unit. ${}^{14}_6\text{C} \longrightarrow {}^{14}_7\text{N} + {}^0_{-1}\text{e}$ Students are not required to recall these two examples. The emission of a gamma ray does not cause the mass or the charge of the nucleus to change.	WS 1.2, MS 1b, 1c, 3c Refer to a copy of the periodic table and use the names and symbols of common nuclei and particles to write balanced equations that show single alpha (α) and beta (β) decay. This includes balancing atomic numbers and mass numbers.

4.3.2.3 Half-life

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the concept of half-life and how this is related to the random nature of radioactive decay.	Radioactive decay is random, so it is not possible to predict which individual nucleus will decay next. But with a large enough number of nuclei it is possible to predict how many will decay in a certain amount of time. The half-life of a radioactive isotope is the average time it takes for the number of nuclei of the isotope in a sample to halve, or the average time it takes for the count rate from a sample containing a radioactive isotope to fall to half its initial level. Count rate is the number of decays recorded each second by a detector (such as a Geiger-Müller tube).	WS 3.3 Carry out and represent mathematical and statistical analysis. MS 4a Determine the half-life of a radioactive isotope from given information. MS 1c, 3d (HT only) Calculate the net decline, expressed as a ratio, in a radioactive emission after a given number of half-lives.

4.3.2.4 Penetration properties of radiations

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Recall the differences in the penetration properties of alpha particles, beta particles and gamma rays.	Alpha particles are absorbed by just a few millimetres of air or by a thin sheet of paper. Beta particles can pass through air and paper but are completely absorbed by a sheet of metal just a few millimetres thick. Gamma rays pass through most materials easily but are absorbed by a thick sheet of lead or by several metres of concrete.	

4.3.2.5 Contamination and irradiation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall the differences between contamination and irradiation effects and compare the hazards associated with these two.	Irradiation is the process of exposing an object to radiation from an outside source. Irradiation can be reduced by screening the source or moving the object away from it. The irradiated object does not become radioactive. Radioactive contamination is the unwanted presence of a source of radiation inside, or on the surface of, other materials. It is often difficult to remove the contaminating source so that it continues to add to the radiation dose for as long as it emits radiation.	

4.3.2.6 Ionising radiations

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Recall that changes in atoms and nuclei can also generate and absorb radiations over a wide frequency range. Describe how ultraviolet waves, X-rays and gamma rays can have hazardous effects, notably on human bodily tissues. Recall that atoms can become ions by loss of outer electrons.	The hazardous effects of ultraviolet (UV) waves, X-rays, alpha, beta and gamma rays depend on the type of radiation and the size of the dose. Radiation dose (in Sieverts) is a measure of the risk of harm resulting from an exposure of the body to the radiation. 1 Sievert (Sv) = 1000 millisieverts (mSv). Ultraviolet waves, X-rays, alpha, beta and gamma rays are all examples of ionising radiation. They can turn atoms into ions and break up molecules. Ionising radiations can change DNA, causing mutation of genes that may lead to cancer. High-energy gamma rays can be used to destroy cancer cells.	WS 1.5 Interpret simple measures of risk showing the probability of harm from different types of radiation. Describe precautions that can be taken to reduce the risks from ionising radiation. Give examples to show that the perceived risk can be very different from the measured risk, especially if the cause of the risk is unfamiliar or invisible.

4.3.2.7 Cancer

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe cancer as the result of changes in cells that lead to uncontrolled growth and division.	Tumours form when cells start growing and dividing in an uncontrolled way. Some tumours are benign; they stay in the same place and stop growing before they get too large. Cancer is caused by malignant tumours that are able to invade neighbouring tissues and spread to different parts of the body in the blood so that more tumours start to grow in other parts of the body.	